



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Time table), this portion gives further specific details regarding the course.

Course No.: PHA G 624

Course Title: Principles of Toxicology

Instructor in charge: Pragyanshu Khare

Tutorial/Practical Instructors: Pragyanshu Khare

1. Course Description:

The course will cover all aspects of the rationale and purpose of preclinical toxicology studies of new drugs under development. It gives an in-depth understanding of the basic principles and mechanisms of toxicology, different classes of toxicants, dose and the concept of the dose-response relationship, Toxicokinetics, and toxicodynamics, OECD guidelines for toxicological study, International Council for Harmonization (ICH) guidelines, toxicity testing methods in organ systems.

2. Scope and Objective of the Course: Students who have undergone the course are expected to

- Understand the molecular mechanisms of toxicity
- Understand the guidelines and methods of testing for toxicity studies
- Understand the regulatory requirements of preclinical toxicological studies
- Understand the ICH and OCED guidelines for preclinical toxicity testing and safety Pharmacology

3. Text Books:

- Casarett & Doull's Toxicology: The Basic Science of Poisons, 9th Edition
- Fundamentals of Toxicology, 1st edition, Elsevier

4. Reference Books:

- 1-Drug Discovery and Evaluation Pharmacological Assays Second Completely Revised, by H. Gerhard Vogel
- 2-The International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH) guidelines



5. Course Plan:

Lect.No.	Module	Lecture Session	References
1-5	Introduction scope and history of toxicology	Introduction to Toxicology, Principles of Toxicology, General Terms used in toxicology-Toxicant/poisons Mechanisms	Textbook Chapter2, Class notes, Text Book Chapter 3
		Types and Classifications of Toxicants. Sub-disciplines of Toxicology, Common Toxic Mechanisms and Sites of Action, Agencies for Toxic Chemical Regulation	
6-12:	Toxicity Testing, Dose-Response, Risk Assessment Toxicity evaluation: methods in organ systems	Toxicological testing: In vivo systems: In vivo tests or bioassays: Drug toxicity mechanisms and methods to detect toxicity In vitro tests: Genotoxicity, Mutagenicity, Chronic, dermal, inhalational, reproductive, and developmental toxicology	Class notes and RB
13-16:	Introduction to ICH and OECD guidelines	Introduction to ICH and OECD guidelines for toxicity testing:	Self-study, official webpages for the recent updates
17-18:	R3 and Replace Animal Use for Toxicity Testing	In vitro assay: EPIDERM TM, EPISKIN TM, Embryonic stem cell test, 3T3 NRU phototoxicity test	Class notes and research papers
19-24	To understand the importance of toxicodynamics toxicokinetics pharmaceutical development	Introduction to toxicodynamics and toxicokinetics and toxicokinetic Concepts	Textbook- Chapter7, ICH guidelines (S3A)
		Disposition of toxicants ADME of Toxicants Biotransformation of Xenobiotics	
25-32	Preclinical toxicological investigations of pharmaceutical and biotechnology products	Drug toxicity mechanisms and methods to detect toxic Responses Regulatory Considerations Specific Poison, Drugs, diagnosis, and treatment Therapeutics drugs and antidote	Textbook chapter 12, ICH guidelines



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33-38	To understand the mechanism and methods of testing for Target Organ Toxicity	Drug toxicity mechanisms and methods to detect toxic Responses of the Liver , Kidney, and Heart	Text book-Chapter 13 and 14, ICH guidelines (S7A and S7B),
39-42	Molecular and Immunochemical Techniques in Toxicology	Molecular Cloning, cDNA, and Genomic Libraries, Northern /Southern Blot Analysis PCR Evaluation of Gene Expression Regulation and Function Proteomics Metabolomics, Bioinformatics	Self-study Class notes and research papers

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Close Book/ Open Book
Mid-Semester Test	90 Min.	30	TBD	CB/OB
Comprehensive Examination	180 Min	35	TBD	CB/OB
Continuous Assessment*	TBD	25	TBD	CB/OB
Class discussions (Linked to attendance)	In class	10	In class	NA

*Based on the theory covered in class. Topics and numbers will be announced in the class. It can be in terms of home assignments, tutorials, projects, laboratory, viva- voce, Quizzes and presentations/ seminars.

Criterion for NC: 30% of the average of the top three performers or 40% of median, whichever is lower, or a suitable modification at the discretion of the IC with the concurrence of the Department

7. Chamber Consultation Hour: To be announced in class.

8. Notices: All the notices pertaining to this course will be displayed only on the Dept. of Pharmacy Notice Board.

9. Make-up Policy: Prior approval or intimation to take a make-up is mandatory. It is solely at the discretion of the instructor-in-charge, depending upon the genuineness of the circumstances, to allow or disallow a student to appear for a make-up evaluation component. No makeup will be granted for Assignments/Quizzes under any circumstances.

Instructor-in-charge

Course No.PHA_G624